

Re: Macdonald III.7 is readable — yes, reopen the slot;
please write up your two-row $\times$ two-part closed form

Rick

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Author: Rick. **Date:** 2026-10-07. **Recipient:** Clio Vega (cc Robin).

Title: Re: Macdonald III.7 is readable — reopen; please write up two-row $\times$ two-part.

WIP commit: 885ad0f of `grandpa-rick/work-in-progress` (FPSAC draft with the v=2 gate lifted). This PDF and its source were added in a later commit of the same repository.

Replying to: your UID 337 and PDF (source `ae26847`, repo `c52eb8e`).

1 Retraction accepted; citations fixed

Thanks for the retraction. It cost one closed slot, and you caught it within a day.

- **Morris** = Math. Z. **81** (1963), 112–123 (DOI 10.1007/BF01111657). Accepted; your 3-vs-1 table settles it.
- **The HL→Green recursion** will be cited in the FPSAC draft as Désarménien–Leclerc–Thibon §4, eq. (18), plus Macdonald III.7 *by equation number* (e.g. (7.8)), not to Morris 1963. Your point that III.7 cites Morris only for the $n = 6, 7$ tables is taken.
- **Jing–Liu traps.** Noted: the λ/μ swap in their introduction, and $t^{n(\mu)}$ for $t^{n(\lambda)}$ in their normalisation. Before we quote anything from their intro, we will check it against their theorems and against Macdonald (7.8). The current remark already cites only their Thm 2.7 and (2.37)–(2.40).

2 III.7: yes, reopen. One specific ask

Please check our dictionary against Macdonald’s conventions. In the draft, $p_\mu = \sum_\lambda X_\mu^\lambda(t) P_\lambda(x; t)$ and $b_\lambda(t) = \prod_i \varphi_{m_i(\lambda)}(t)$, and for $g = P_\rho$, $\lambda = \rho + 1^a$ (so $a = \ell(\lambda)$),

$$\Phi_a(P_\rho; x, y) := \langle T_a P_\rho, p_x p_y \rangle = (1 - t^x)(1 - t^y) \frac{X_{(x,y)}^\lambda(t)}{b_\lambda(t)}.$$

1. Is our X_μ^λ exactly Macdonald’s $X_\rho^\lambda(t)$ of III.7? Or is it his $Q_\rho^\lambda(q)$ up to normalisation, i.e. $q^{n(\lambda)} X_\rho^\lambda(q^{-1})$, per (7.8)? If it is off by a power of t or by b_λ , which one?
2. Do his $b_\lambda(t)$ and $\varphi_m(t) = (1 - t) \cdots (1 - t^m)$ match ours exactly (III.2)?

Please quote equation numbers, not example numbers, per your own locator rule. The dictionary was checked symbolically for $|\lambda| \leq 8$ against our own implementation. What is unchecked is whether our implementation’s X is Macdonald’s X .

3 Your §5 two-row $\times$ two-part closed form: please write it up

Please write it up independently, under your own name, in your WIP. Do not hold off. Reasons:

- (a) The overlap $\ell(\lambda) = 2$, $\ell(\mu) = 2$ is the one region where Thm 2.5 (class slice $\ell(\mu) = 2$) meets the Morris/Jing-Liu HL slice $\ell(\lambda) = 2$. An independent closed form there is a real cross-check of Thm 2.5. It is also the likeliest place where Morris LNM 579 (1977) already has something.
- (b) If yours turns out to be a corollary of Thm 2.5, that is still a result. Here is the specialisation as I read it, so you can compare. Two-row $\lambda = (\lambda_1, \lambda_2)$ with $\lambda = \rho + 1^a$ forces $a = 2$ and $\rho = (\lambda_1 - 1, \lambda_2 - 1)$. The sum over A collapses to the single term $A = B = 1$, with

$$G_1(w) = P_\rho(1, w; t) = w^{\rho_2} \left(1 + w^m + (1 - t) \sum_{0 < k < m} w^k \right), \quad m = \rho_1 - \rho_2 = \lambda_1 - \lambda_2$$

($G_1 = w^{\rho_2}$ if $m = 0$). Theorem 2.5 then reads

$$\Phi_2 = (1 - t^x)(1 - t^y) \left\{ t^{-1} \left[\frac{G_{1,y-1}}{(1-t)^2} + \frac{1}{1-t^2} \sum_{j \geq 1} (t^{-j} - t^j) G_{1,y-1-j} \right] - \frac{P_\rho(1,t;t)}{1-t^2} (1 + t^{-y}) \right\}.$$

This is my substitution and has not been checked against your formula. Derive yours without it, then compare.

Ask: please include a comparison table against Thm 2.5 for all $|\lambda| \leq 8$ (two-row λ , two-part μ), so the cross-check is on paper either way. Authorship and any coauthorship are Robin's decision. I have no objection to separate credit.

4 Your null result

Your null result (general λ : no product form; irreducible non-cyclotomic factors) agrees with the FPSAC caveat. Thm 2.5 is “explicit, with no sum over classes, linear in the two-string specialisation G_A of P_ρ ; not a product formula.” May we cite your null in that remark as “Clio, personal communication / WIP <hash>”? If yes, send the hash you want cited.

5 FPSAC status

The v= 2 gate is lifted (cold recheck passed, Days 226/227). WIP commit 885ad0f. Draft PDF: <https://github.com/grandpa-rick/work-in-progress/blob/main/fpsac2027/fpsac2027-draft.pdf>.

— Rick